

# Abstract

This project aims to design and implement a comprehensive cryptographic system that explores both classical and modern encryption techniques using Python. It covers a wide range of algorithms including Caesar Cipher, Playfair Cipher, Hill Cipher, Vigenère Cipher, Simplified Data Encryption Standard (SDES), Advanced Encryption Standard (AES), Stream Cipher (RC4), Diffie–Hellman Key Exchange, and RSA. Each algorithm is implemented and tested to understand its working principles, strengths, and limitations.

The system follows a client–server architecture to facilitate secure communication over a network. The client encrypts the message using the selected algorithm and sends it to the server. The server decrypts the message and displays the original content, and it can also send messages back to the client using the same process, demonstrating two-way secure communication.

Classical algorithms provide insight into the early methods of encryption such as substitution and transposition, while modern algorithms like AES, RC4, and RSA showcase the power of key-based and asymmetric cryptography. Diffie–Hellman enables both parties to securely establish a shared secret key before communication. The PyCryptodome library is used to implement and support the modern cryptographic algorithms.

This project serves as a practical learning platform for understanding the fundamentals of cryptography and their real-world applications in ensuring confidentiality, integrity, authentication, and secure data exchange. The implementation demonstrates how different cryptographic techniques can be integrated into a single system to build secure and reliable communication between networked systems.